

Component design

```
templates/index.html
├──
└── ▼
    POST /generate-workout
    ├──
    ├── ── UserInput validation
    ├── ── Gemini workout generator
    ├── ── Gemini nutrition generator
    └── ── SQLAlchemy / SQLite
├──
├──
├── ▼
├── result.html
├──
├──
├── ▼
├── POST /submit-feedback
├──
├──
├── ▼
├── Gemini update layer
├──
├──
├── ▼
├── updated_plan
```

Data model

users table:

- id
- user_id
- username
- age
- weight
- goal
- intensity
- original_plan
- updated_plan
- nutrition_tip
- created_at
- updated_at

AI model responsibilities

- Pro-capability configurable model: structured workout generation and feedback updates.
- Flash-capability configurable model: quick nutrition/recovery tip.

The actual model IDs are environment variables rather than hard-coded architectural assumptions.

Security design

- API key is server-side only.
- `.env` is ignored by Git.
- Admin route requires a configured local demo key.
- User input is validated before AI/database processing.